3rd YEAR PORTFOLIO PROJECT — VIT PUNE CSE-DS

ZOMATO RESTAURANT INTELLIGENCE

A Complete, Nine-Phase Data Analytics & Deployment Project

9	41,410	R^2 0.82	LIVE
Project Phases	Restaurants Analyzed	ML Model Accuracy	Deployed Web Application

Yash Chavan | B.Tech CSE-DS, VIT Pune | Batch 2024-28
Live App · zomato-growth-navigator.streamlit.app | **GitHub** · github.com/yashtc2239-ops/zomato-restaurant-intelligence

01 OVERVIEW
Executive Summary

This report documents a complete, nine-phase data analytics project analyzing **41,410+ Zomato restaurants** across Bengaluru, supplemented by a 6,513-restaurant multi-city Indian subset from a second global dataset. The project moves from raw, uncleaned CSV files through business framing, ETL, exploratory analysis, SQL querying, Power BI dashboarding, machine learning, and ends in a **fully deployed, publicly accessible Streamlit web application**. Every phase was driven by a specific business question rather than generic exploration — the goal throughout was decision-grade analysis, not a classroom exercise.

Project Roadmap

#	Phase	Key Output
1	Business Understanding	Problem statement, 7 MECE business questions
2	Dataset Understanding	Schema audit of 2 Kaggle datasets (61K+ rows)
3	Data Cleaning (ETL)	41,410 + 6,513 production-clean restaurant rows
4	Exploratory Data Analysis	6 business questions answered statistically
5	Visualization	5 business-driven, insight-labeled charts
6	SQL Analysis	8 business queries — 2 linked SQL tables
7	Power BI Dashboard	3-page dashboard, 5 custom DAX measures
8	Advanced Business Analysis	Expansion Score algorithm + Random Forest ML
9	Deployment & Engineering	Streamlit + Docker + live Cloud deployment

PHASE 01 · 02

BUSINESS & DATASET UNDERSTANDING

1.1 PHASE 1 Business Understanding

Zomato was framed as a three-sided marketplace connecting restaurants (supply), customers (demand), and the platform itself. Rather than producing generic restaurant statistics, the project was scoped around a single, decision-oriented business problem:

> ***"Using Zomato restaurant data, identify the key drivers of customer ratings and build a pricing/performance benchmark that helps the business development team prioritize restaurant partnerships in underserved segments."***

Seven business questions were defined across four MECE themes:

Theme	Business Questions
Demand & Customer Behavior	Cuisine vs order volume by city tier; price vs rating correlation
Supply & Restaurant Performance	Restaurant density by city; attributes that predict higher ratings
Operational Efficiency	Delivery time by city/cuisine/time; online order + booking combo
Strategic / BI	Which cities show high demand but low supply — where to expand

1.2 PHASE 2 Dataset Understanding

Two public Kaggle datasets were audited and combined to give both **depth** (city-level granularity) and **breadth** (multi-city comparison) — something neither dataset offered alone.

Dataset	Rows	Cols	Role in Project
Zomato Bengaluru (himanshupoddar)	51,717	17	PRIMARY — used in every phase: cleaning, EDA, SQL, Power BI, ML model, and the deploy
Zomato Global (shrutimehta)	9,551	21	SECONDARY — filtered to India (6,513 rows), used in Phase 2 schema audit, Phase 4 EDA

Audit findings: zero duplicate rows in either dataset. Bengaluru's *rate* column was stored as a string fraction (e.g. "4.1/5") with inconsistent spacing and placeholder values ("NEW", "-"). The *cost_for_two* column used comma-separated thousands ("1,200"). *dish_liked* was 54.3% null and was dropped. The Global dataset was 90.5% India records — filtering to India removed currency inconsistency with minimal data loss.

PHASE 03 · 04

DATA CLEANING & EXPLORATORY ANALYSIS

1.3 PHASE 3 Data Cleaning (ETL)

Issue	Decision	Reason
rate stored as "4.1/5"	Regex extract to float	Enables numeric aggregation
dish_liked 54% null	Drop column	Too sparse to be reliable
cost has commas ("1,200")	Strip + cast to float	Required for price-bracket analysis
online_order/book_table Yes/No	Map to 1/0	Needed for correlation & ML model
Global: 90.5% India rows	Filter Country Code == 1	Removes currency inconsistency
Useless columns (url, phone, etc.)	Dropped	Zero analytical value

Result: Bengaluru reduced from 51,717 → 41,410 rows (17 → 11 columns). Global (India) reduced from 9,551 → 6,513 rows (21 → 14 columns), standardized to match the Bengaluru column naming convention for direct comparison.

1.4 PHASE 4 Exploratory Data Analysis

Six business questions were answered statistically on the cleaned Bengaluru dataset:

Question	Finding
Rating distribution	43.6% in Good (3.5-4.0) tier; only 1.4% reach Excellent (4.5+)
Online order vs rating	Negligible — 3.72 vs 3.66 avg (0.06 point difference)
Cost vs rating	Pearson r = 0.38 (moderate) — Luxury 4.16 vs Budget 3.57
Location vs rating	Lavelle Road leads at 4.14; Bommanahalli trails at 3.19
Cuisine popularity vs quality	North Indian dominates supply but ranks outside top 10 by rating
Table booking vs rating	Strongest predictor — 4.14 vs 3.62 avg, 2.6x higher cost

★ Cross-dataset insight: the Global (India) subset showed a lower average rating (3.35 vs 3.70) and higher average cost (■716 vs ■603) — confirming Bengaluru as a relatively higher-rated, lower-cost market within India's broader restaurant landscape.

PHASE 05 · 06

VISUALIZATION & SQL ANALYSIS

1.5 PHASE 5 Visualization

Five matplotlib/seaborn charts were produced, each answering one explicit business question rather than being decorative. Every chart carries a one-line business insight as a subtitle — the "so what?" principle applied throughout.

Chart File	Business Question Answered
chart1_rating_distribution.png	How are restaurants distributed across rating tiers?
chart2_table_booking.png	Does table booking indicate a premium segment?
chart3_cost_vs_rating.png	Does spending more guarantee a better experience?
chart4_location_rating.png	Which locations should Zomato prioritize or improve?
chart5_cuisine_analysis.png	Which cuisines are oversupplied vs underserved?

1.6 PHASE 6 SQL Analysis

Both cleaned datasets were loaded into SQLite as two tables — *bengaluru* and *global_india* — and queried with 8 production-style business queries (also available as MySQL-compatible syntax).

Query	Business Use
Q1 — Restaurant type distribution	Understand Zomato's product/listing mix
Q2 — Top 10 locations by rating	Target areas for premium brand partnerships
Q3 — Online order impact on rating	Confirms online ordering alone doesn't drive quality
Q4 — Table booking premium analysis	Identify the premium restaurant tier
Q5 — Cuisine performance	Surface underserved high-quality cuisine segments
Q6 — Demand/supply ratio by location	Identify expansion-ready vs saturated markets
Q7 — Restaurant type performance	Which restaurant formats perform best
Q8 — Expansion recommendation	Final ranked list of HIGH/MEDIUM/SATURATED zones

★ Hero finding (Q6/Q8): Church Street and Lavelle Road show 1,000+ votes per restaurant — more than double the city average — flagging both as demand-starved, expansion-ready zones.

PHASE 07

POWER BI DASHBOARD

A 3-page interactive Power BI dashboard was built on the cleaned Bengaluru dataset, using 5 custom DAX measures to move beyond simple SUM/COUNT aggregations into comparative and relative-performance metrics — the kind of work that separates a tutorial project from an industry-level deliverable.

Dashboard Pages

Page	Contents
1 · Executive Summary	4 KPI cards, rating distribution, order/booking donuts, cost-bracket slicer
2 · Location Intelligence	Top/Bottom 10 locations, cost-vs-rating scatter, Expansion Priority table
3 · Cuisine & Business	Supply vs Quality comparison, table-booking cards, Value Score rankings

DAX Measures

Measure	Purpose
City Avg Rating	Context-independent baseline using ALL() for fair comparisons
Rating Variance vs City	Shows how far each location deviates from the city baseline
Value Score	Price-tier-adjusted rating using ALLEXCEPT — surfaces hidden gems
Performance Flag	Conditional Top Performer / Underperformer tag
Demand Pressure Index	Votes-per-restaurant relative to city average — the expansion signal

DAX measures using **ALL()** and **ALLEXCEPT()** filter-context manipulation are a common differentiator between tutorial-level and industry-level Power BI work — demonstrating fluency beyond drag-and-drop visuals.

PHASE 08

ADVANCED BUSINESS ANALYSIS

Expansion Score Algorithm

Expansion Score = (40% × Demand Pressure) + (35% × Avg Rating) + (25% × Supply Gap)

All three metrics normalized 0–100 via MinMaxScaler across 72 locations with 30+ restaurants. **3 zones classified Critical priority** (score > 80).

Rank	Location	Score	Priority	Rest.	Rating
1	Lavelle Road	95.6	CRITICAL	481	4.14
2	Church Street	91.1	CRITICAL	546	3.99
3	St. Marks Road	81.6	CRITICAL	343	4.02
4	Koramangala 3rd Block	78.1	HIGH	191	4.02
5	Koramangala 4th Block	76.2	HIGH	841	3.92

Rating Prediction Model

Model	R^2 Score	RMSE	Verdict
Linear Regression	0.2920	0.3696	Weak — linear assumption insufficient
Random Forest	**0.8157**	**0.1885**	**WINNER — 179% better R^2**

Feature	Importance	Interpretation
votes (engagement)	69.3%	**DOMINANT — 4.7x more important than price**
cost_for_two	14.7%	Moderate influence on quality
rest_type	10.9%	Format matters — fine dining rates highest
online_order	3.0%	Minimal direct impact
book_table	2.1%	Confirms premium segment signal

★ Key Finding: Customer engagement (votes) is 4.7x more important than price in predicting restaurant quality — Zomato's ranking algorithm should weight engagement signals higher than price-based factors.

PHASE 09

DEPLOYMENT & ENGINEERING

The final phase converts the analysis into a production-style deployed product rather than leaving it as static notebooks — the difference between a data analyst portfolio and an engineer's portfolio.

Streamlit Web Application — Live & Public

Page	Features
Executive Overview	4 live KPI cards, rating distribution, service donuts, cost-vs-quality chart
Location Intelligence	Top/Bottom 10 locations, color-coded Expansion Priority table (live-filtered)
Cuisine & Business	Supply vs Quality charts, table-booking premium signal comparison
ML Model Insights	Feature importance, model comparison table, Live Rating Predictor widget

Sidebar filters (Location, Cuisine, Price Range, Min Rating) dynamically re-filter all four pages in real time, including re-computation of the Expansion Score and ML feature charts. Edge cases — such as a filter combination returning zero or near-zero restaurants — are handled gracefully with informative messages rather than runtime errors.

Containerization (Docker)

Layer	Purpose
FROM python:3.11-slim	Pinned, lightweight base image for reproducibility
WORKDIR /app	Isolated, clean working directory inside the container
COPY requirements.txt + RUN pip install	Dependency layer cached separately from app code
COPY . .	Application code layer — changes most frequently
EXPOSE 8501	Documents the Streamlit port for cloud routing
HEALTHCHECK	Lets cloud platforms detect a crashed container
CMD streamlit run ...	Entry point — binds to 0.0.0.0 for external access

Layer ordering follows Docker best practice: dependencies are installed before application code is copied in, so code edits do not invalidate the (slow) pip install cache layer.

Cloud Deployment — Continuous Delivery

The application is deployed on Streamlit Community Cloud, connected directly to the GitHub repository. Every git push to the main branch triggers an automatic rebuild and redeploy — a lightweight CI/CD pipeline with zero manual deployment steps after initial setup.

Item	Detail
Live URL	zomato-growth-navigator.streamlit.app
Source repo	github.com/yashtc2239-ops/zomato-restaurant-intelligence
Deployment model	Streamlit Cloud — auto-redeploy on every git push

Local alternative	Docker — docker build + docker run, identical to cloud behavior

FINAL OUTPUT

BUSINESS RECOMMENDATIONS & DELIVERABLES

1 **EXPAND in 3 Critical Priority Zones**

Lavelle Road (95.6), Church Street (91.1), St. Marks Road (81.6) — high demand pressure, strong ratings, relatively low supply. Onboard new restaurant partners here before saturation.

2 **REDESIGN the Rating Algorithm**

Customer votes drive 69% of rating prediction — 4.7x more than cost. Reweight engagement signals in the ranking algorithm to improve recommendation quality.

3 **TARGET the Premium Cuisine Gap**

Modern Indian (4.31), European (4.26), Mediterranean (4.20) outperform in ratings but have low restaurant counts — high demand, low competition opportunity.

4 **SCALE the Table Booking Program**

Table booking restaurants rate 0.52 points higher and charge 2.6x more (■1,276 vs ■482). Expanding Zomato Gold/Pro captures meaningful revenue upside.

5 **QUALITY IMPROVEMENT for Bottom 10 Locations**

Bommanahalli (3.19), RT Nagar (3.46) have high restaurant counts but poor quality scores. Targeted training programs would lift overall platform rating.

02 DELIVERABLES Project Outputs

Deliverable	Description
Live Streamlit App	4-page interactive dashboard, deployed on Streamlit Cloud
GitHub Repository	Full source code, notebooks, SQL, reports, Docker config
Power BI Dashboard (.pbix)	3-page dashboard with 5 custom DAX measures
6 Jupyter Notebooks	Phases 1-8 fully documented, reproducible analysis
SQL Business Queries	8 queries covering Bengaluru + Global India tables
Dockerfile	7-layer container definition, production-ready
5 Chart PNGs + 2 ML charts	Expansion score, feature importance, and 5 EDA visualizations
This Report (PDF)	Full 9-phase business and technical analysis